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OOP LAB

Assignment #01

Program #01

1. Find Maximum and Minimum number in Array.

```
#include <iostream>
using namespace std;
int main( ){
int n; numbers[100];
 int min;
int max;
cout << "Enter number of element";
cin >> n;
for (int i = 0; i <n; i++){
cout << "Enter number" << i+1 << " ";
cin >> number[i];

 max = number[0];
min = number[0];
for (int i = 1; i<n; i++){
if (number[i] > max){

 max = number[i];
}
```

```
if (number[i] < min)
    min = number[i];
  }
}
cout << "Maximum number " << max << endl;
cout << "Minimum number" << min << endl;
return 0;
}
```

Output

```
Enter number of element : 2

Enter 1 number : 3
Enter 2 number : 2

Maximum number = 3
Minimum number = 2.

program executed
```

Program #02

Vehicle Registration System

```
#include <iostream>
using namespace std;
struct Vehicle {

string regNO;
string ownerName;
```

```
string model;
int year;
};
int main ()
int n;
cout << "Enter number of vehicles";
cin >> n;

vehicles v[100];

for (int i=0 ; i<n ; i++)
cout << "\n vehicle" << i+1 << endl;
cout << "Registration Number";

cin >> v[i].regNo;
cout << "Owner name";
cin >> v[i].owner name;
cout << "Model";
cin >> v[i].model;

cout << "Manufacturing Year";
cin >> v[i].year;

}
cout << "\n vehicle Records";
for (int i=0;i<n;i++)

cout << "\n Registration Number" << v[i].regNo;
cout << "Owner name" << v[i].owner name << endl;
cout << "Model" << v[i].model << endl;
cout << " Year" << v[i].year << endl;
}
```

```
return 0;
}
```

Output.

Registration Number : 25008

owner Name : BILAL Abbasi

Manufacturing year : 2025

vehicle: Alto765.

Program #03.

Employee Payroll System. Dynamic Allocation.

```
#include<iostream>
using namespace std;
int main(){

  int n;
cout<<"Enter number of employee";
cin>>n;
double *salary =new double[n];

for(int i=0;i<n;i++){
```

(2)

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```
for (int i=0 ;i<n ;i++)
cout << "Enter sallary of employee
        << (i+1) << endl;
  cin >> sallary [i];
  }
  for (int i=0 ;i<n ;i++) {
  cout << "Employee" << i+1 << "salary" <<
      salary[i] << endl;
  }
    delete [] salary;
      return 0;
  }
```

Output

```
Enter number of salary : 2
salary of employee 1 : 25000
salary of employee 2 : 35000
```

Program #04.

Library Book Management System.

```
#include <iostream>
Using namespace std;
  class Book {
   private:

   string tittle;
   String author;
   string ISBN;

   string availibility status;

      public:

void set Fittle (string t) {title=t;}
void set Author (string a)
                  {author = a;}.

void set ISBN (string i) {isBN=i;}


void set status (string s)
                  {availibility status=s;}
```

```
string get tittle(){return tittle;}
string get Author(){return Author;}
string get ISBN(){return ISBN}
string get status(){return status;}
};
    int main(){
Book books[3];
for(int i=0;i<3;i++){
cout<<"\n Enter Book"<<i+1<<
                    Details\n";
  string<<"Tittle";
   cin>>tittle;
cout<<"Author";
  cin>> Author;
   cout<<"ISBN";
    cin>> ISBN;
books[i].set Title(tittle);
books[i].set Author(author);
books[i].set ISBN(isbn);
```

```
cout << "\n Library books";
for (int i=0 ; i<3 ;i++) {
 cout << "\n Book" << i+1 << endl;
 cout << " Title" << book[i].getTitle()
cout << "ISBN" << book[i].getISBN() << endl;
cout << "Status" << book[i].getStatus() << endl;
 }
    return 0;
}
```

Output.

```
ISBN: 2
Enter book 2 Details.

Enter title: Programming.
Author: RAJA ALI
 ISBN: 4

Enter book 3 Details:

Enter tittle: CALCULUS..
Author: Mustafa.
 ISBN: 6.
```

Program 05

CAR

```
#include <iostream>
using namespace std;

class car {
  private:
  int car ID;
 string model;
 double rent per Day;
bool is Avalible;

  public:        // Default.
 car() {
  car ID = 0;
  model = "Unknown";
 rent per Day = 0.0;
  is Avalible = true;
}
    // parameterized

car(int id, string m, double rent,
      bool status) {

  car ID = id;
  model = m;
 rent per day = rent;
is Avalible = status;
```

```
// copy constructor
car(const &car &c){
    carID = c.carID;
   model = c.model;

 rentperDay = c.rentperDay;
isAvalible = c.isAvalible;
}  // destructor

~Car(){
cout << "Car with ID" << carID
     << removed << endl;

  // Rent Car

void rentCar(){
 if(isAvalible){

   isAvalible = false;

cout << "car rent successfully" <<
                    endl;

 else{

 cout << "car is already rented"
      << endl;
```

```
// Display Details
void displayCarDetails(){
cout<<"\n Car ID: "<< carID <<endl;
cout<<"\n Model "<< model <<endl;
cout<<"\n Rent per day "<< rentPerDay
                        <<endl;
 cout<<" Availibility "<<(isAvalible)
 (Avalible : Rented) << endl;
}
};
 int main(){
  Car c1;
  // Parameterized.
Car c2(101,"Toyota Corolla",500,true)
        // copy object.
   Car c3=c2;
c2.rentCar();
c2.displayCarDetails();
   // Display all cars
```

```
cout << "\n Car c1 Details";
c1.display car Details();
cout << "\n Car 3 Details";
c3.display car Details();
return 0;
}
```

Output.

```
Car ID : 101
Model : Toyota corolla
Rent per Day : 5000
Availibility : Rented.
Car rented successfully.

Car ID : 202
Model : Honda
Rent per Day : 10000
Availibility : Rented

Car Rented successfully.
car with ID 101 removed from system.
car with 0 removed from
system.
```